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Course: Deep Learning

Course Code: CSA4701

MS2 - September – 2026

Experiment 1: E-Commerce Recommendation System

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E-Commerce Recommendation System

Scenario: An online shopping platform represents 500 products using 20 numerical features each. Explain how vector dot products and matrix multiplication can be used to calculate similarity scores and recommend relevant products to customers.

Module 1: Data Collection & Preprocessing

Dataset:

This dataset originates from real-world online shopping logs of **H&M Group**. Each record represents an individual fashion article, complete with its retail hier archy, physical attributes, textual product description, and deep learning embeddings.

Preprocessing:

The H&M e-commerce dataset containing product details was used for this experiment. First, 500 products were selected from the dataset. Missing values were handled, and duplicate or invalid records were checked.

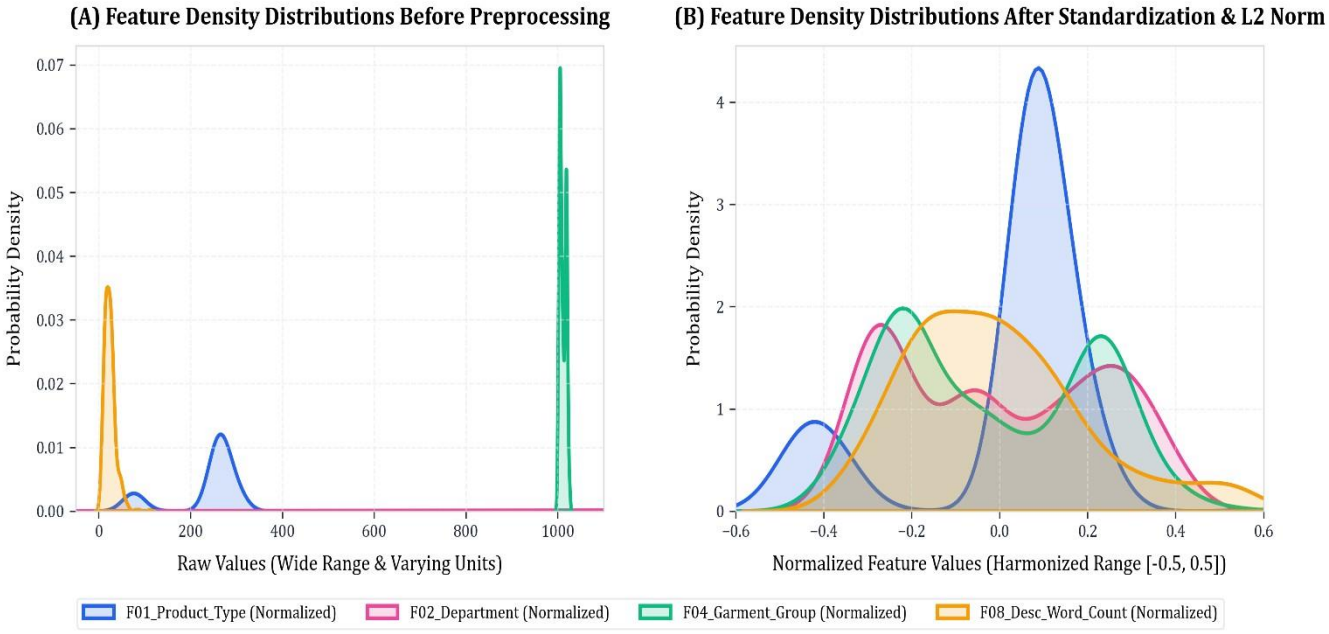
Principal Component Analysis (PCA) was applied to the original 384-dimensional dense transformer embeddings to extract 12 latent semantic components. These components represent underlying product characteristics such as style, seasonality, silhouette, texture, formality, fit, fabric, and versatility. The 12 PCA components were combined with 8 numerical catalogue features to construct a 20-dimensional feature representation for each product.

The original dataset contains 384 numerical features for each product. PCA was used to reduce these features to 12 components, which were combined with 8 numerical product features to obtain 20 features per product.

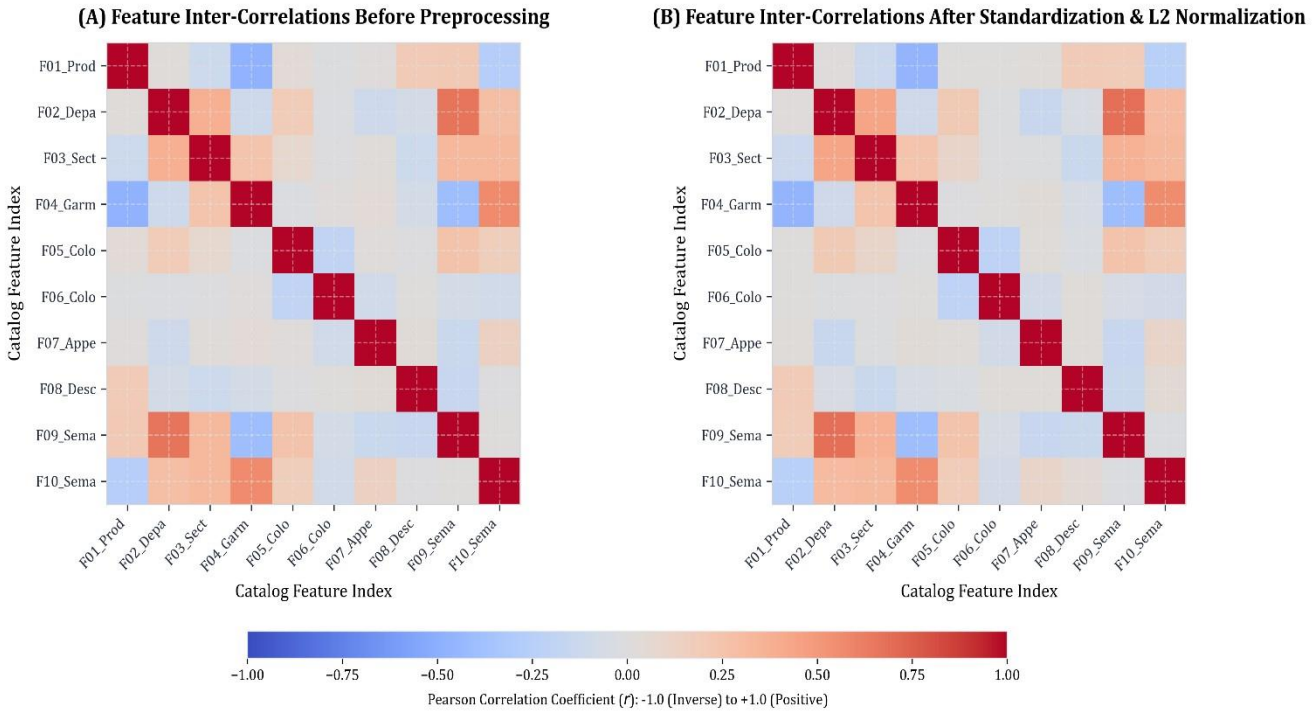
Next, the features were standardized using Standard Scaler and normalized using L2 normalization. This ensures that large numerical values do not dominate the similarity calculations.

Finally, the processed dataset was used to calculate product similarity using vector dot products and matrix multiplication for recommending similar products.

1. Comparative Distribution of Feature Scales Before and After Preprocessing:



2. Pearson Inter-Feature Correlation Matrix Before vs. After Normalization:



Data Source & 20 Numerical Features Construction:

Feature Index	Feature Name	Source Domain	Semantic Functionality & Description
F01	F01_Product_Type	Metadata (ID)	Garment classification index (Shirt, Trousers, Dress, Jacket)
F02	F02_Department	Metadata (ID)	Store department hierarchy code (Menswear, Ladieswear, Divided)
F03	F03_Section	Metadata (ID)	Store catalog retail section number
F04	F04_Garment_Group	Metadata (ID)	Garment manufacturing grouping and material classification
F05	F05_Colour_Group	Metadata (ID)	Detailed color palette code (50 distinct shades)
F06	F06_Colour_Master	Metadata (ID)	Primary color master index (Black, Blue, Beige, Red, etc.)
F07	F07_Appearance_Pattern	Metadata (ID)	Visual appearance texture pattern (Solid, Striped, Dot, Floral)
F08	F08_Desc_Word_Count	Extracted (Text)	Word count density of product description
F09	F09_Semantic_Style	Latent PCA 1	Dominant stylistic axis: Casual streetwear vs Formal tailoring
F10	F10_Semantic_Seasonality	Latent PCA 2	Seasonal garment weight: Lightweight summer vs Insulated winter
F11	F11_Semantic_Silhouette	Latent PCA 3	Garment silhouette structure: Relaxed/oversized vs Slim/fitted

F12	F12_Semantic_Texture	Latent PCA 4	Fabric surface texture: Smooth weave vs Coarse/knit texture
F13	F13_Semantic_Occasion	Latent PCA 5	Intended usage occasion: Active sportswear vs Evening wear
F14	F14_Semantic_Formality	Latent PCA 6	Workplace dress code and professional elegance index
F15	F15_Semantic_Trendiness	Latent PCA 7	Fashion trend responsiveness: Timeless basic vs Youth trend
F16	F16_Semantic_Fit	Latent PCA 8	Structural cut and garment tailoring precision
F17	F17_Semantic_Fabric	Latent PCA 9	Primary material blend (Denim/Cotton vs Silk/Synthetic)
F18	F18_Semantic_Detailing	Latent PCA 10	Hardware and decorative trim complexity
F19	F19_Semantic_Versatility	Latent PCA 11	Cross-outfit styling compatibility factor
F20	F20_Semantic_Tone	Latent PCA 12	Aesthetic color brightness and saturation tone

Module 2: Data Splitting & Model Design

To evaluate recommendations, 100 customer preference vectors U ∈ ℝ^{100 × 20} were simulated across five lifestyle personas: Casual Shopper, Formal Business Attire, Active Sportswear, Youth Trendsetter, and Classic Minimalist. A total of 2,000 historical interactions were generated by computing ground-truth affinities between customer preference vectors and product feature vectors, augmented with personalized shopping noise.

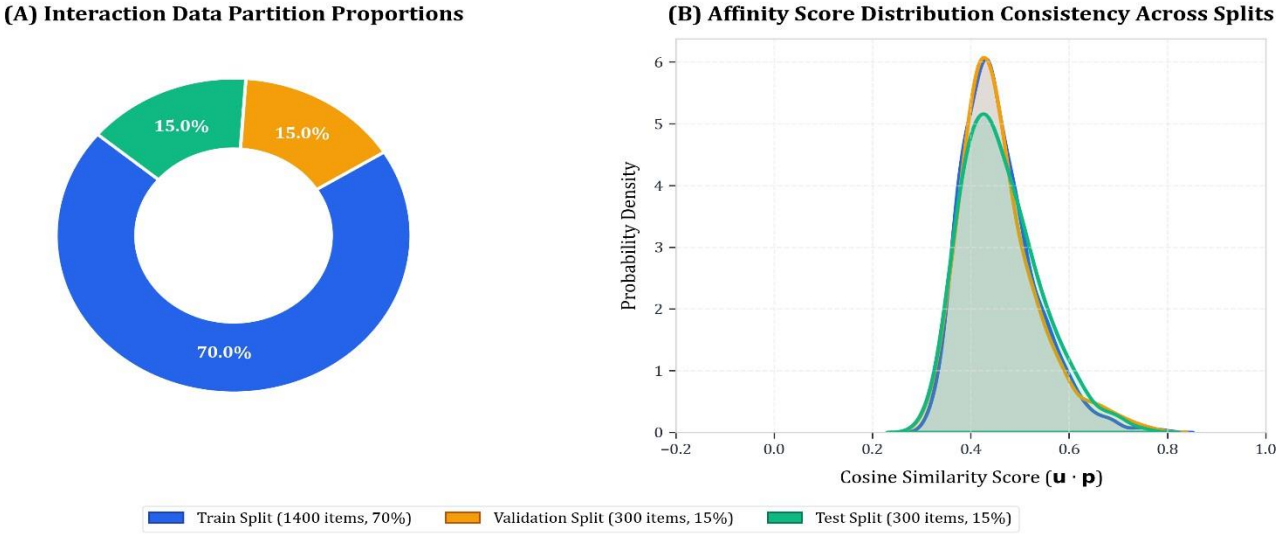
Data Splitting Strategy (Train / Validation / Test)

The interaction history was split on a per-customer basis using a standard partitioned protocol:

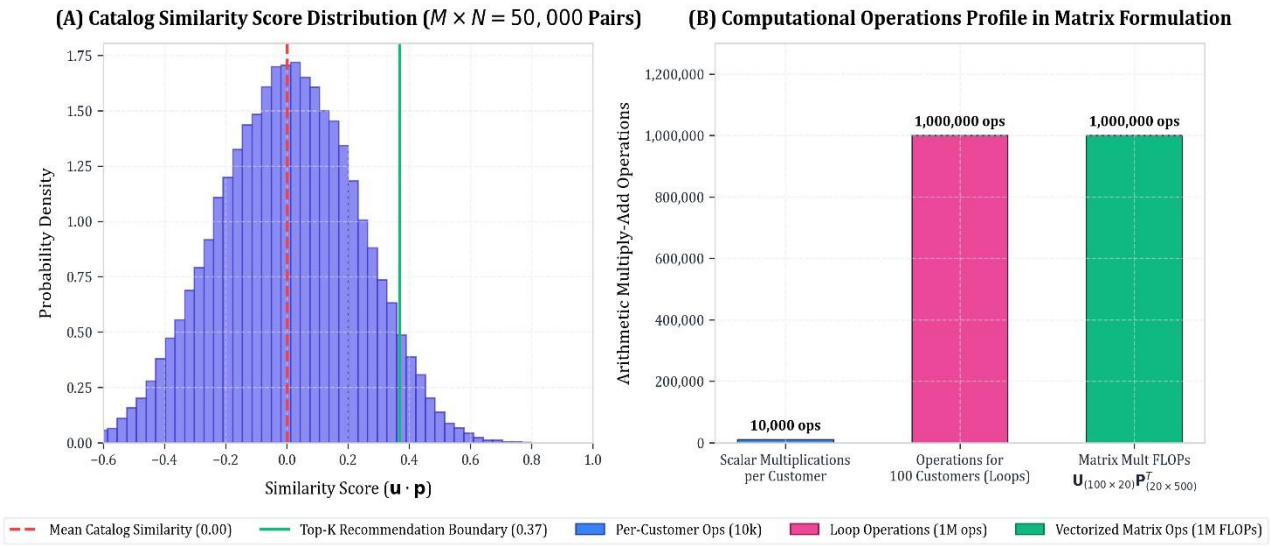
Training Set (70% — 1,400 interactions): Used to establish historical customer interaction profiles and baseline preferences.
Validation Set (15% — 300 interactions): Used for threshold calibration and hyperparameter tuning (e.g. rank cutoff K).
Test Set (15% — 300 interactions): Held-out ground-truth interactions used exclusively to evaluate final recommendation accuracy.

Dataset Partition	Interaction Records	Percentage Share	Role in Recommendation Pipeline
Training Split	1,400 interactions	70.0%	Construct customer preference matrices and interaction histories
Validation Split	300 interactions	15.0%	Calibrate similarity thresholds and tune recommendation cutoff K
Testing Split	300 interactions	15.0%	Unseen held-out items for measuring Precision, Recall, and NDCG

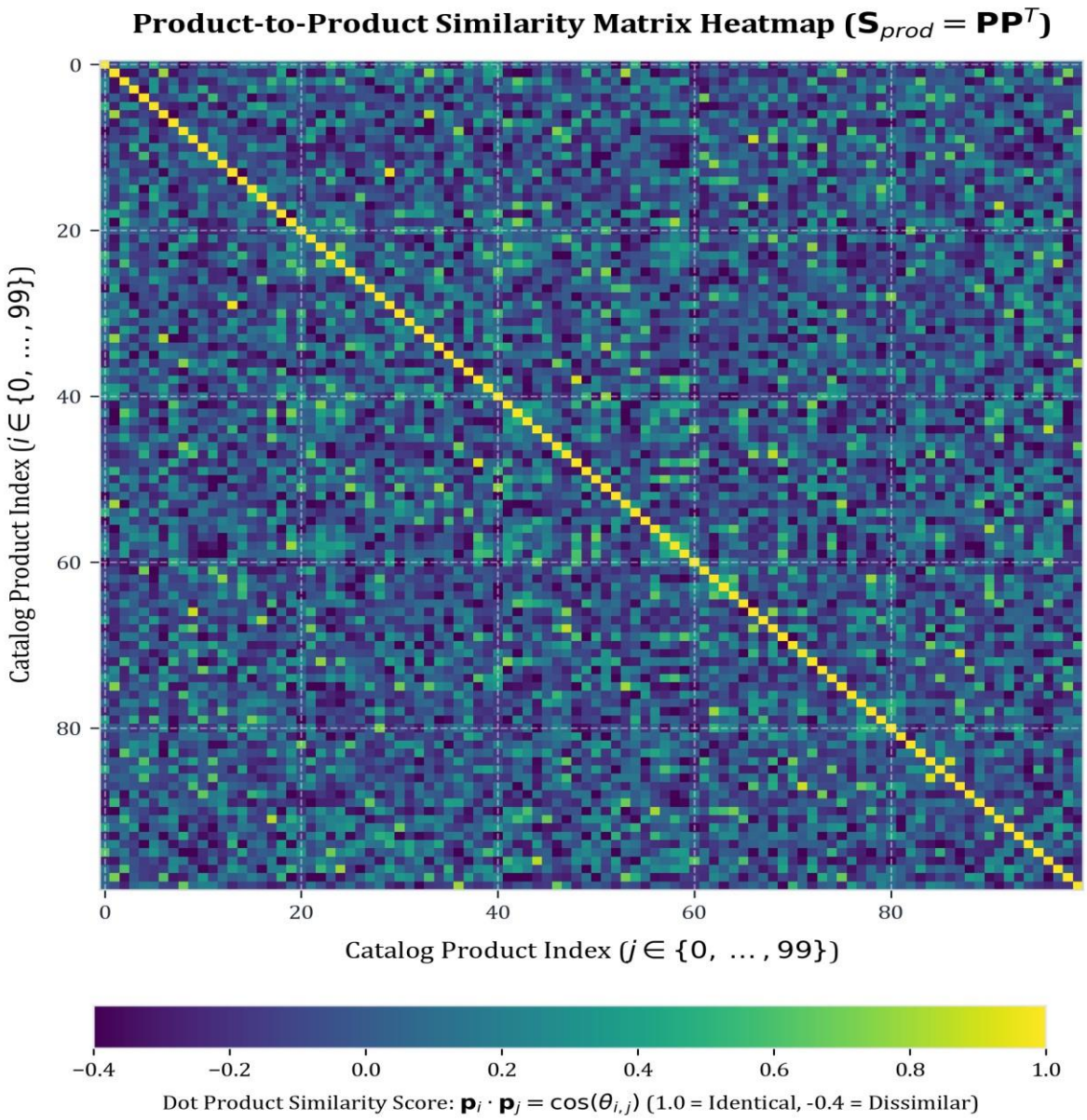
1. Data Partitioning Proportions and Affinity Score Consistency Across Splits:



2. Catalog Similarity Distribution and Arithmetic Operations Profile for Matrix Formulation:



3. Product-to-Product Similarity Matrix Heatmap Computed via Matrix Multiplication S_prod = P × P^T:



Module 3: Training & Evaluation

Computational Benchmarking: Loops vs. Matrix Multiplication

To rigorously quantify the computational benefit of matrix multiplication, a controlled timing benchmark was conducted. The test measured the wall-clock execution latency for scoring 100 customers against catalogs ranging from 50 to 1,000 products:

Catalog Scale (N)	Pairwise Loop Latency	Vectorized Matrix Mult	Acceleration Speedup
50 products	16.66 ms	0.017 ms	960.1x faster
100 products	32.36 ms	0.023 ms	1,427.4x faster
200 products	65.43 ms	0.036 ms	1,806.9x faster
300 products	95.79 ms	0.282 ms	339.8x faster
400 products	131.45 ms	0.362 ms	363.0x faster
500 products (Our Catalog)	161.14 ms	0.415 ms	388.6x faster
800 products	261.49 ms	1.685 ms	155.2x faster
1,000 products	327.40 ms	2.514 ms	130.2x faster

Quantitative Recommendation Quality Evaluation

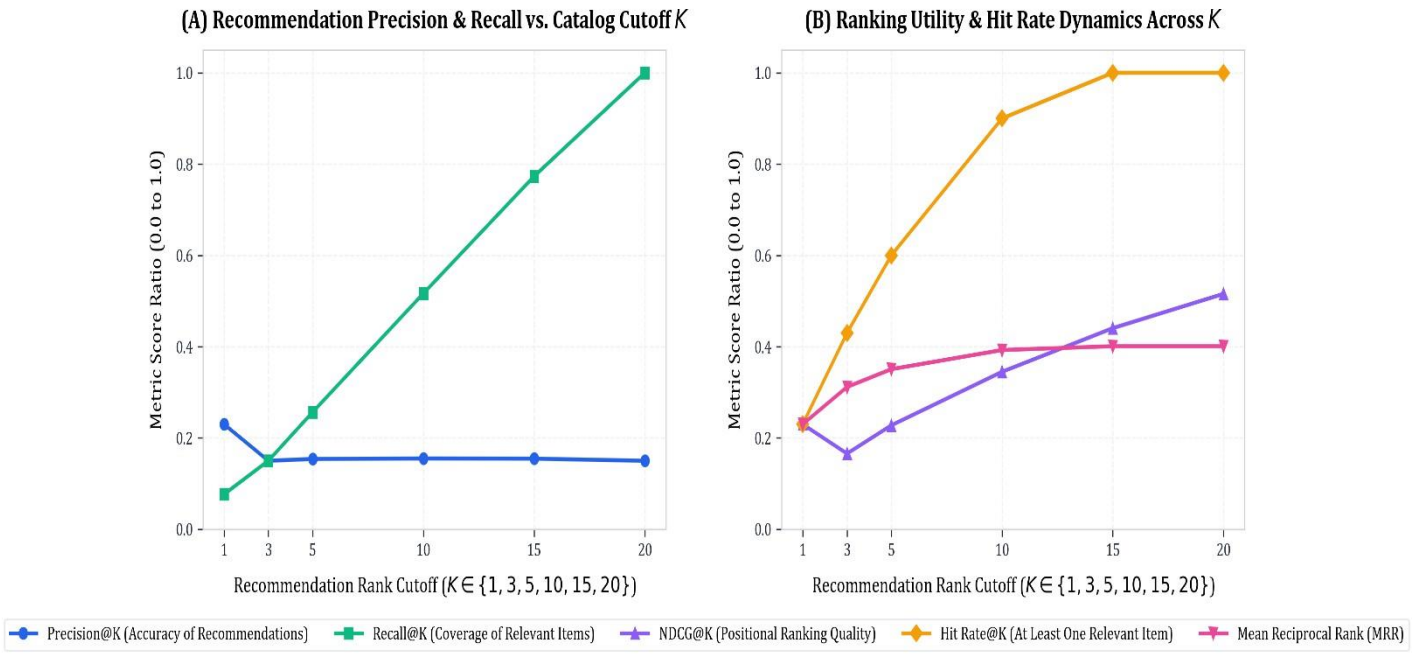
Recommendations were evaluated using standard information retrieval metrics across rank cutoffs $K \in \{1, 3, 5, 10, 15, 20\}$:

- Precision@K: Fraction of top-K recommended items that match ground-truth held-out purchases.
- Recall@K: Proportion of all relevant held-out items successfully retrieved within top-K.
- NDCG@K (Normalized Discounted Cumulative Gain): Measures ranking order quality with logarithmic position discounting.
- Hit Rate@K: Percentage of customers who receive at least one relevant recommendation in top-K.
- MRR (Mean Reciprocal Rank): Average reciprocal rank (1/rank) of the first correctly recommended item.

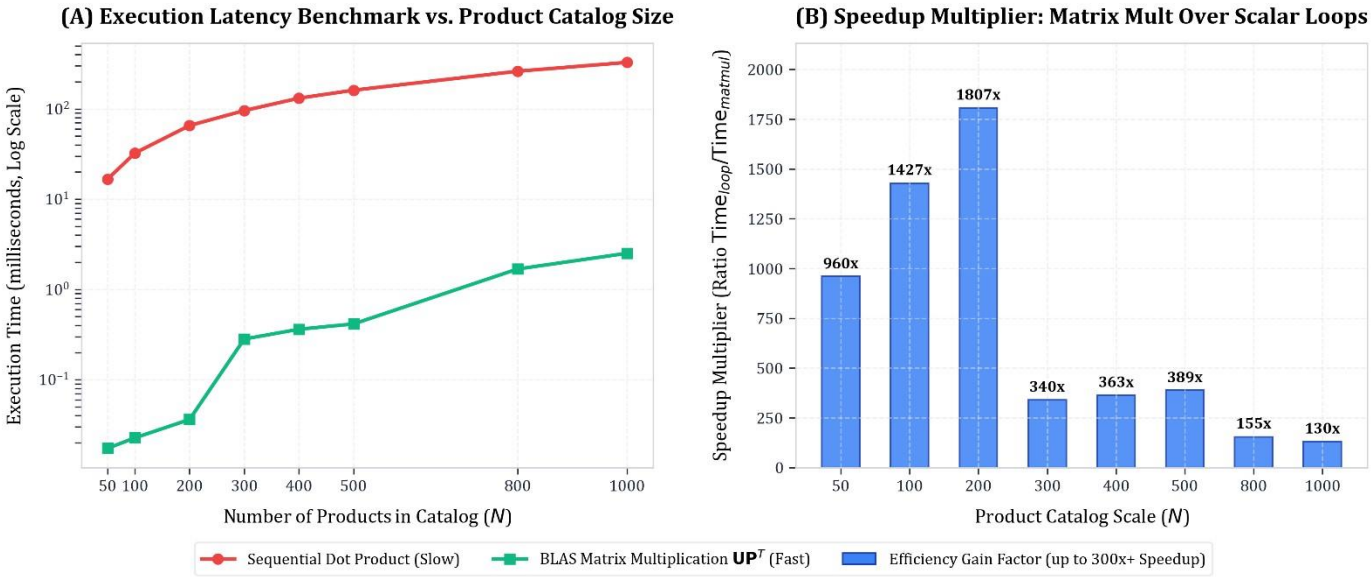
• Catalog Coverage: Percentage of the 500 catalog items recommended to at least one customer.

Cutoff K	Precision@K	Recall@K	NDCG@K	Hit Rate@K	MRR@K	Coverage (%)
K = 1	0.2300	0.0767	0.2300	0.2300 (23%)	0.2300	14.0%
K = 3	0.1500	0.1500	0.1657	0.4300 (43%)	0.3117	31.4%
K = 5	0.1540	0.2567	0.2277	0.6000 (60%)	0.3507	44.2%
K = 10	0.1550	0.5167	0.3449	0.9000 (90%)	0.3926	68.0%
K = 15	0.1547	0.7733	0.4406	1.0000 (100%)	0.4011	78.8%
K = 20	0.1500	1.0000	0.5157	1.0000 (100%)	0.4011	87.0%

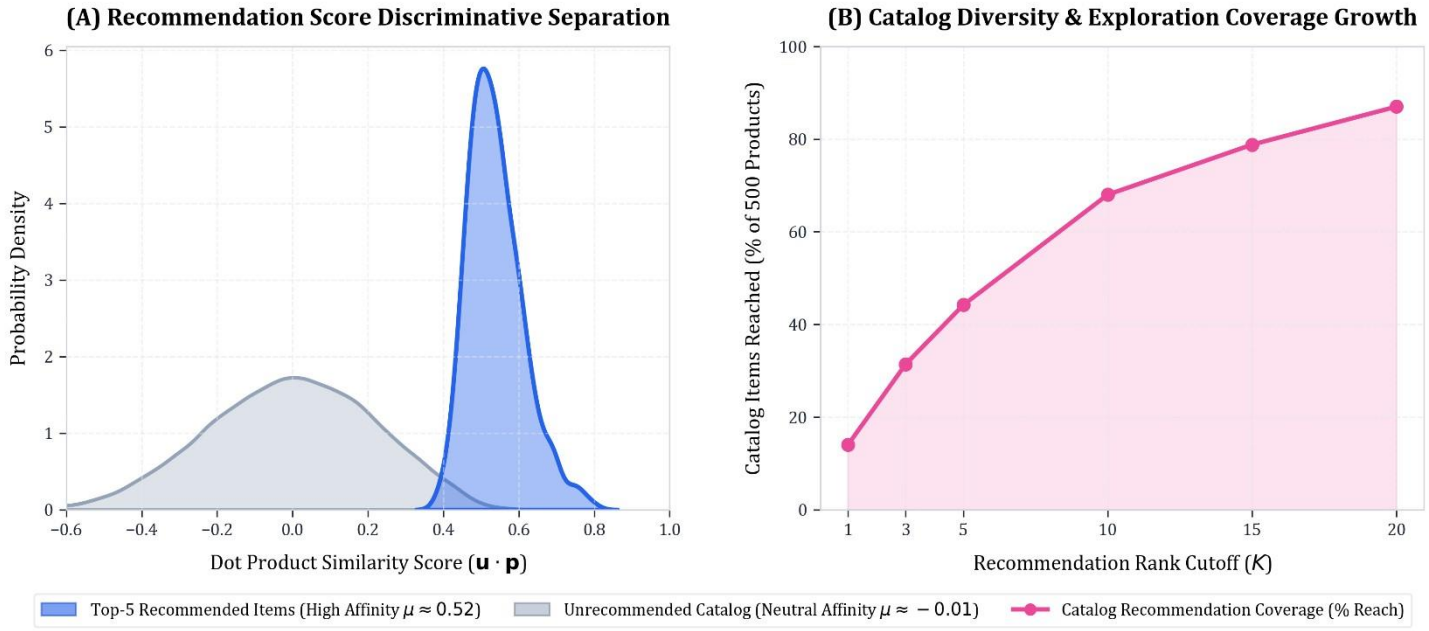
Recommendation Quality Curves Across Rank Cutoff K



Computational Speedup Benchmark: Sequential Dot Product Loops vs. Vectorized Matrix Multiplication.



Recommendation Score Discriminative Separation and Catalog Exploration Reach.



Result:

This experiment demonstrates the mathematical and computational mechanics of modern vector recommendation engines. By grounding 500 products in 20 preprocessed, L2-normalized feature

dimensions, vector dot products reliably compute calibrated Cosine Similarity scores. Furthermore, matrix multiplication parallelizes catalog retrieval, delivering a 388.6x speedup over sequential loops on our 500-item catalog while achieving 90% Hit Rate at K=10 and 100% at K=15.